

高等微积分

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第六章: 求导的逆运算-不定积分





6.3. 有理函数的不定积分

设 $Q(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ ($a_n \neq 0$).
由代数基本定理可知 Q 有 n 个根 (包括重数),
其中复根成对出现. 于是

$$Q(x) = a_n \prod_{j=1}^s (x - \alpha_j)^{l_j} \cdot \prod_{k=1}^t (x^2 + p_kx + q_k)^{m_k},$$

这里 $\alpha_j \in \mathbb{R}$ 均不相同, $p_k^2 - 4q_k < 0$, 而且

$$\sum_{j=1}^s l_j + 2 \sum_{k=1}^t m_k = n.$$

任意的有理真分式 $R(x) = \frac{P(x)}{Q(x)}$ (P, Q 为多项式
且 $\deg P < \deg Q$) 可分解为

$$R(x) = \sum_{j=1}^s \sum_{u=1}^{l_j} \frac{a_{j,u}}{(x - \alpha_j)^u} + \sum_{k=1}^t \sum_{v=1}^{m_k} \frac{b_{k,v}x + c_{k,v}}{(x^2 + p_kx + q_k)^v},$$

其中 $a_{j,u}, b_{k,v}, c_{k,v} \in \mathbb{R}$ 为常数. 为证明分解式,
我们可将 $a_{j,u}, b_{k,v}, c_{k,v}$ 看成未知元 (共有 n 个),
两边乘以 $Q(x)$, 比较多项式的系数, 得到 n 个
线性方程, 由此可以唯一确定未定元的值.

于是有理真分式 $R(x) = \frac{P(x)}{Q(x)}$ 最终可以分解成如下 4 种最简单的分式之和 ($m \geq 2$):

$$\frac{A}{x-\alpha}, \quad \frac{A}{(x-\alpha)^m}, \quad \frac{Ax+B}{x^2+px+q}, \quad \frac{Ax+B}{(x^2+px+q)^m} \quad (p^2 - 4q < 0).$$

由于 $x^2 + px + q = (x + \frac{p}{2})^2 + \frac{4q-p^2}{4}$, 经过变量替换, 有理分式的不定积分可归结成下述 6 种最简单的分式的不定积分 ($a > 0$):

$$\frac{1}{x-\alpha}, \quad \frac{1}{(x-\alpha)^m}, \quad \frac{x}{x^2+a^2}, \quad \frac{1}{x^2+a^2}, \quad \frac{x}{(x^2+a^2)^m}, \quad \frac{1}{(x^2+a^2)^m}.$$

这些不定积分有显式表达式:

- $\int \frac{dx}{x-\alpha} = \ln |x - \alpha| + C,$
- $\int \frac{dx}{(x-\alpha)^m} = -\frac{1}{(m-1)(x-\alpha)^{m-1}} + C,$
- $\int \frac{x dx}{x^2+a^2} = \frac{1}{2} \ln(x^2 + a^2) + C,$

- $I_1 = \int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan \frac{x}{a} + C,$
- $\int \frac{x dx}{(x^2+a^2)^m} = -\frac{1}{2(m-1)} \frac{1}{(x^2+a^2)^{m-1}} + C,$
- $I_{m+1} = \int \frac{dx}{(x^2+a^2)^{m+1}} = \frac{x}{2a^2 m (x^2+a^2)^m} + \frac{2m-1}{2a^2 m} I_m.$

例 36. 计算 $\int \frac{2x^2+2x+13}{(x-2)(x^2+1)^2} dx$.

解: 由题设可知 $\frac{2x^2+2x+13}{(x-2)(x^2+1)^2}$ 的标准分解形如

$$\frac{2x^2+2x+13}{(x-2)(x^2+1)^2} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}.$$

将上式两边同乘以 $(x-2)(x^2+1)^2$ 后可得

$$\begin{aligned} 2x^2 + 2x + 13 &= (A+B)x^4 + (C-2B)x^3 \\ &+ (2A+B-2C+D)x^2 + (C-2B+E-2D)x \\ &+ (A-2C-2E). \end{aligned}$$

比较系数并解方程组可得

$$A = 1, B = -1, C = -2, D = -3, E = -4.$$

故 $\frac{2x^2+2x+13}{(x-2)(x^2+1)^2} = \frac{1}{x-2} + \frac{-x-2}{x^2+1} + \frac{-3x-4}{(x^2+1)^2}$, 由此可得

$$\begin{aligned} \int \frac{2x^2 + 2x + 13}{(x-2)(x^2+1)^2} dx &= \int \frac{dx}{x-2} - \int \frac{x dx}{x^2+1} \\ &\quad - 2 \int \frac{dx}{x^2+1} - 3 \int \frac{x dx}{(x^2+1)^2} - 4 \int \frac{dx}{(x^2+1)^2} \\ &= \ln|x-2| - \frac{1}{2} \ln(x^2+1) - 2 \arctan x + \frac{3}{2} \frac{1}{x^2+1} \\ &\quad - 4 \left(\frac{x}{2(x^2+1)} + \frac{1}{2} \arctan x \right) + C \\ &= \frac{1}{2} \ln \frac{(x-2)^2}{x^2+1} - 4 \arctan x + \frac{12-4x}{2x^2+1} + C. \end{aligned}$$

例 37. 计算 $\int \frac{x^4-2x^3+3x^2-x+1}{x^4-3x^3+3x^2-x} dx$.

解: 由带余除法可得

$$\frac{x^4 - 2x^3 + 3x^2 - x + 1}{x^4 - 3x^3 + 3x^2 - x} = 1 + \frac{x^3 + 1}{x^4 - 3x^3 + 3x^2 - x}.$$

又 $x^4 - 3x^3 + 3x^2 - x = x(x-1)^3$, 由此我们知

$\frac{x^3+1}{x^4-3x^3+3x^2-x}$ 的标准分解形如:

$$\frac{x^3 + 1}{x^4 - 3x^3 + 3x^2 - x} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2} + \frac{D}{(x-1)^3}.$$

两边同乘以 $x^4 - 3x^3 + 3x^2 - x$ 可得

$$\begin{aligned}x^3 + 1 &= (A + B)x^3 + (-3A - 2B + C)x^2 \\ &\quad + (3A + B - C + D)x - A.\end{aligned}$$

由此可得 $A = -1, B = 2, C = 1, D = 2$. 则

$$\begin{aligned}\int \frac{x^4 - 2x^3 + 3x^2 - x + 1}{x^4 - 3x^3 + 3x^2 - x} dx &= \int dx - \int \frac{dx}{x} \\ &\quad + 2 \int \frac{dx}{x-1} + \int \frac{dx}{(x-1)^2} + 2 \int \frac{dx}{(x-1)^3} \\ &= x - \ln|x| + 2 \ln|x-1| - \frac{1}{x-1} - \frac{1}{(x-1)^2} + C.\end{aligned}$$

例 38. 计算 $\int \frac{2x}{(x+1)(x^2+1)^2} dx$.

解: 由题设可知 $\frac{2x}{(x+1)(x^2+1)^2}$ 的标准分解形如:

$$\frac{2x}{(x+1)(x^2+1)^2} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}.$$

两边乘以 $(x+1)(x^2+1)^2$ 可得

$$\begin{aligned} 2x &= A(x^2+1)^2 + (Bx+C)(x+1)(x^2+1) \\ &\quad + (Dx+E)(x+1). \end{aligned}$$

在上式中选取 $x = -1$, 由此立刻可得 $A = -\frac{1}{2}$.

比较系数可得 $B = \frac{1}{2}$, $C = -\frac{1}{2}$, $D = E = 1$, 则

$$\frac{2x}{(x+1)(x^2+1)^2} = \frac{-1}{2(x+1)} + \frac{x-1}{2(x^2+1)} + \frac{x+1}{(x^2+1)^2},$$

$$\begin{aligned} \int \frac{2x}{(x+1)(x^2+1)^2} dx &= \int \frac{-1}{2(x+1)} dx + \int \frac{x}{2(x^2+1)} dx \\ &+ \int \frac{-1}{2(x^2+1)} dx + \int \frac{x}{(x^2+1)^2} dx + \int \frac{1}{(x^2+1)^2} dx \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2} \ln |x+1| + \frac{1}{4} \ln(x^2+1) - \frac{1}{2} \arctan x \\
&\quad - \frac{1}{2(x^2+1)} + \left(\frac{x}{2(x^2+1)} + \frac{1}{2} \arctan x \right) + C \\
&= \frac{1}{4} \ln \frac{x^2+1}{(x+1)^2} + \frac{x-1}{2(x^2+1)} + C.
\end{aligned}$$

例 39. 计算 $\int \frac{x^2+1}{(x+1)(x-2)^2} dx$.

解: 由题设可知 $\frac{x^2+1}{(x+1)(x-2)^2}$ 的标准分解形如

$$\begin{aligned} \frac{x^2+1}{(x+1)(x-2)^2} &= \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2} \\ &= \frac{A(x-2)^2 + B(x+1)(x-2) + C(x+1)}{(x+1)(x-2)^2}. \end{aligned}$$

故 $x^2+1 = A(x-2)^2 + B(x+1)(x-2) + C(x+1)$.

令 $x = -1$ 可得 $A = \frac{2}{9}$. 令 $x = 2$ 则可得 $C = \frac{5}{3}$.
再取 $x = 0$ 可得 $B = \frac{7}{9}$. 于是我们有

$$\frac{x^2+1}{(x+1)(x-2)^2} = \frac{2}{9(x+1)} + \frac{7}{9(x-2)} + \frac{5}{3(x-2)^2}.$$

由此立刻可得

$$\begin{aligned} \int \frac{x^2 + 1}{(x+1)(x-2)^2} dx &= \int \frac{2}{9(x+1)} dx \\ &+ \int \frac{7}{9(x-2)} dx + \int \frac{5}{3(x-2)^2} dx \\ &= \frac{2}{9} \ln |x+1| + \frac{7}{9} \ln |x-2| - \frac{5}{3(x-2)} + C. \end{aligned}$$



§ 6.4. 可有理化函数的不定积分

假设 $R(u, v) = \frac{P(u, v)}{Q(u, v)}$, 其中 $P(u, v), Q(u, v)$ 均为关于变量 u, v 的多项式. 所谓的三角有理函数就是指 $R(\sin x, \cos x)$. 下面我们将由万能公式来将之转化成有理分式的不定积分.

令 $t = \tan \frac{x}{2}$, 则 $x = 2 \arctan t$. 于是

$$dx = d(2 \arctan t) = \frac{2}{1+t^2} dt,$$

$$\sin x = \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}} = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{2t}{1+t^2},$$

$$\cos x = \frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}} = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{1-t^2}{1+t^2}.$$

从而我们有

$$\int R(\sin x, \cos x) \, dx \stackrel{t=\tan \frac{x}{2}}{=} \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2}{1+t^2} \, dt.$$

例 40. 计算 $\int \frac{1+\sin x}{1+\cos x} dx$.

解:

$$\begin{aligned}\int \frac{1+\sin x}{1+\cos x} dx &\stackrel{t=\tan \frac{x}{2}}{=} \int \frac{1+\frac{2t}{1+t^2}}{1+\frac{1-t^2}{1+t^2}} \frac{2}{1+t^2} dt = \int \frac{1+t^2+2t}{1+t^2} dt \\&= \int \left(1 + \frac{2t}{1+t^2}\right) dt = t + \ln(1+t^2) + C \\&= \tan \frac{x}{2} + \ln(1+\tan^2 \frac{x}{2}) + C = \tan \frac{x}{2} - 2 \ln |\cos \frac{x}{2}| + C.\end{aligned}$$

在一些特殊情形, 上述讨论可以简化:

- 被积函数为 $\sin x$ 的奇函数 (将 $\sin x$ 变换成 $-\sin x$ 后会出现一个负号):

$$\int R(\sin^2 x, \cos x) \sin x dx \stackrel{t=\cos x}{=} - \int R(1-t^2, t) dt.$$

- 被积函数为关于 $\cos x$ 的奇函数:

$$\int R(\sin x, \cos^2 x) \cos x dx \stackrel{t=\sin x}{=} \int R(t, 1-t^2) dt.$$

例 41. 计算 $\int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x} \quad (ab \neq 0).$

解:
$$\begin{aligned} & \int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x} \\ &= \int \frac{1}{a^2 \tan^2 x + b^2} \cdot \frac{dx}{\cos^2 x} \\ &= \int \frac{d(\tan x)}{a^2 \tan^2 x + b^2} = \int \frac{d\left(\frac{a}{b} \tan x\right)}{ab\left(1 + \left(\frac{a \tan x}{b}\right)^2\right)} \\ &= \frac{1}{ab} \arctan \left(\frac{a}{b} \tan x\right) + C. \end{aligned}$$

例 42. 计算 $\int \frac{\cos^3 x}{\sin^7 x} dx$.

$$\begin{aligned}\text{解: } \int \frac{\cos^3 x}{\sin^7 x} dx &= \int \frac{\cos^2 x}{\sin^7 x} d(\sin x) \stackrel{t=\sin x}{=} \int \frac{1-t^2}{t^7} dt \\ &= \int (t^{-7} - t^{-5}) dt = -\frac{1}{6t^6} + \frac{1}{4t^4} + C \\ &= -\frac{1}{6\sin^6 x} + \frac{1}{4\sin^4 x} + C.\end{aligned}$$

例 43. 计算 $\int \sec^3 x \, dx$.

解: $\int \sec^3 x \, dx = \int \sec x \, d(\tan x)$

$$= \sec x \cdot \tan x - \int \tan x \, d(\sec x)$$

$$= \sec x \cdot \tan x - \int \frac{\sin x}{\cos x} \cdot \frac{\sin x}{\cos^2 x} \, dx$$

$$= \sec x \cdot \tan x - \int (\sec^3 x - \sec x) \, dx$$

$$= \sec x \cdot \tan x + \ln |\sec x + \tan x| - \int \sec^3 x \, dx.$$

某些无理函数的不定积分

考虑不定积分 $\int R(x, y(x)) dx$, 其中 $y = y(x)$ 为无理函数, $R(x, y)$ 是关于变量 x, y 的有理分式. 我们希望寻求变量替换 $x = x(t)$ 来使得原来的不定积分能转化成以 t 为变量、前面已解决的不定积分. 常见的情形有以下两种.

$$1. y(x) = \sqrt[n]{\frac{ax+b}{cx+d}}, n \geq 1 \text{ 为整数}, ad - bc \neq 0.$$

解: 令 $t = \sqrt[n]{\frac{ax+b}{cx+d}}$, 则 $t^n = \frac{ax+b}{cx+d}$, 从而 $x = \frac{dt^n - b}{a - ct^n}$,
于是 $dx = \frac{n(ad-bc)t^{n-1}}{(a-ct^n)^2} dt$, 进而可得

$$\int R\left(x, \sqrt[n]{\frac{ax+b}{cx+d}}\right) dx$$
$$\stackrel{t=\sqrt[n]{\frac{ax+b}{cx+d}}}{=} \int R\left(\frac{dt^n - b}{a - ct^n}, t\right) \frac{n(ad-bc)t^{n-1}}{(a-ct^n)^2} dt.$$

例 44. 计算 $\int \frac{dx}{\sqrt[3]{(x-1)(x+1)^2}}.$

解: $\int \frac{dx}{\sqrt[3]{(x-1)(x+1)^2}} = \int \sqrt[3]{\frac{x+1}{x-1}} \frac{1}{x+1} dx$

$$\stackrel{t=\sqrt[3]{\frac{x+1}{x-1}}}{=} \int t \cdot \frac{1}{-\frac{1+t^3}{1-t^3}+1} d\left(-\frac{1+t^3}{1-t^3}\right) = \int \frac{1-t^3}{-2t^2} \cdot \left(-\frac{6t^2}{(1-t^3)^2}\right) dt$$

$$= \int \frac{3}{(1-t)(1+t+t^2)} dt = \int \left(\frac{1}{1-t} + \frac{t+2}{1+t+t^2}\right) dt$$

$$= -\ln|1-t| + \int \frac{(t+\frac{1}{2})+\frac{3}{2}}{\frac{3}{4}+(t+\frac{1}{2})^2} dt$$

$$2. y(x) = \sqrt{ax^2 + bx + c}, a \neq 0.$$

解: 通常先将 $ax^2 + bx + c$ 配方, 然后再来应用三角函数将原来那个不定积分转化成三角有理函数的不定积分.

例 45. 计算 $\int \frac{dx}{2 + \sqrt{x^2 - 2x + 5}}.$

解: $\int \frac{dx}{2 + \sqrt{x^2 - 2x + 5}} = \int \frac{dx}{2 + \sqrt{(x-1)^2 + 4}}$

$$\begin{aligned} \xrightarrow{x=1+2\tan t} \int \frac{d(1+2\tan t)}{2 + \sqrt{4\tan^2 t + 4}} &= \int \frac{\frac{2}{\cos^2 t} dt}{2 + \frac{2}{\cos t}} = \int \frac{dt}{(\cos t) \cdot (1 + \cos t)} \\ |t| < \frac{\pi}{2} \end{aligned}$$

$$= \int \frac{dt}{(\cos t) \cdot (1 + \cos t)} = \int \left(\frac{1}{\cos t} - \frac{1}{1 + \cos t} \right) dt$$

$$= \ln |\sec t + \tan t| - \tan \frac{t}{2} + C_1.$$

由于 $\tan t = \frac{x-1}{2}$, $\sec t = \sqrt{1 + \tan^2 t} = \frac{1}{2}\sqrt{x^2 - 2x + 5}$,

$\tan \frac{t}{2} = \frac{1 - \cos t}{\sin t} = \frac{\sec t - 1}{\tan t} = \frac{\sqrt{x^2 - 2x + 5} - 2}{x - 1}$, 于是我们有

$$\int \frac{dx}{2 + \sqrt{x^2 - 2x + 5}} = \ln |\sec t + \tan t| - \tan \frac{t}{2} + C_1$$

$$= \ln(\sqrt{x^2 - 2x + 5} + x - 1) - \frac{\sqrt{x^2 - 2x + 5} - 2}{x - 1} + C.$$





同学们辛苦了!